

Sampling and Variability

Project: Verified Study Buddy

1. Your mission

Your learning target

I can sample from a distribution using explicit cumulative intervals.

I can calculate temperature-adjusted probabilities for a toy example.

I can explain why random variation and factual reliability need separate evaluation.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

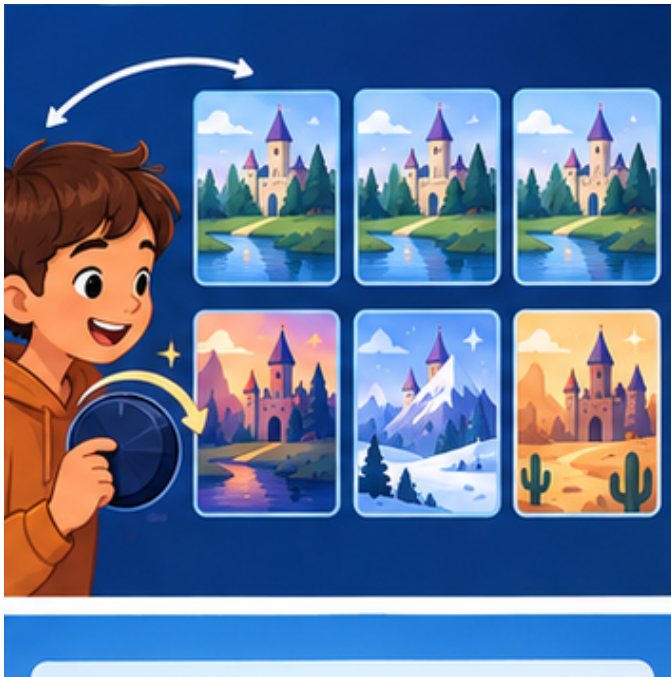
Without opening the old lesson, explain one key idea from Tokens and Context. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Predictive and Generative AI. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 4: Sampling and Variability

Decoding choices trade consistency, diversity, and risk.



CREATOR MISSION

Compare low- and high-variability story continuations.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Does the largest next-token probability guarantee the largest complete-path probability?

Readiness for A small attention calculation

What must happen to permitted weights after excluding a position?

My readiness explanation

This week’s end goal

By the end, I can calculate a toy temperature change and trace a sampling draw.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use sampling and variability to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Start with probabilities red 0.6, blue 0.3, green 0.1 in that fixed order. Greedy chooses red. Sampling uses a draw u with $0 \leq u < 1$ and cumulative intervals: red $[0,0.6)$, blue $[0.6,0.9)$, green $[0.9,1)$. Left endpoints are included, right endpoints excluded. Thus $u=0.6$ selects blue and $u=0.9$ selects green. The draw 1 is outside this contract.

Sampling does not mean choosing all tokens equally. Across many independent draws the frequencies may approach the specified probabilities, but a small sample need not match them exactly. Fixed recorded draws allow a reproducible classroom trace. A random seed and generator version help replay a real sampling program, not establish the truth of outputs.

Apply the connected idea: A small attention calculation

For one query q and keys k_i of width d , compute score $s_i = \text{dot}(q, k_i) / \sqrt{d}$. Then obtain weights from softmax and combine value vectors. Dot products may be negative; probabilities cannot. The softmax performs the conversion. This is a single-head classroom calculation, not a full language model.

Define softmax weights as $\exp(s_i - m) / \sum_j \exp(s_j - m)$, where m is the maximum permitted score. Subtracting the same maximum from every permitted score leaves the ratios unchanged and avoids unnecessary overflow from large positive scores. Natural exponent $\exp$ is introduced here; use a calculator or Python `math.exp` rather than estimating its meaning from the word exponential.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



IN

INPUT

Compare low- and high-variability story continuations.



SYSTEM

Decoding choices trade consistency, diversity, and risk.



OUT

EVIDENCE

Compare low- and high-variability story continuations.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

For strictly positive probabilities p_i and temperature $T > 0$, define $q_i = p_i^{(1/T)} / \sum_j p_j^{(1/T)}$. This classroom formula is equivalent to scaling logits before a softmax when p came from those logits. We use it directly on the supplied table. At $T=1$ the distribution is unchanged. At $T=0.5$, square each probability: 0.36, 0.09, 0.01; total 0.46. The adjusted distribution is red 18/23, blue 9/46, green 1/46.

Token · T=1 probability · T=0.5 probability

Token: red

T=1 probability: 0.6

T=0.5 probability: $18/23 \approx 0.782609$

Token: blue

T=1 probability: 0.3

T=0.5 probability: $9/46 \approx 0.195652$

Token: green

T=1 probability: 0.1

T=0.5 probability: $1/46 \approx 0.021739$

Keep the evidence roles separate

Plan or input · Procedure · Permitted conclusion

Plan or input: $T = 1$

Procedure: Intervals 0 to .6 to .9 to 1

Permitted conclusion: Draw .7 selects blue

Plan or input: $T = 0.5$

Procedure: First boundary 18/23

Permitted conclusion: Draw .7 selects red

Plan or input: Same recorded draw

Procedure: Changed distribution

Permitted conclusion: Different output possible

Temperature changes probability concentration; the same draw can fall in a different interval.

Apply the connected idea: A small attention calculation

Item · Key, width 1 · Value, width 1

Item: A

Key, width 1: [0]

Value, width 1: [2]

Item: B

Key, width 1: $[\ln(3)]$

Value, width 1: $[6]$

Take $q=[1]$ and $d=1$. The scale $\sqrt{1}$ is 1, so scores are 0 and $\ln(3)$. $\ln(3)$ is the number whose natural exponential is 3, approximately 1.098612. Exponentials have ratio 1:3, hence weights $1/4$ and $3/4$. Output is $(1/4)*2 + (3/4)*6 = 5$. Using maximum subtraction gives exponentials $1/3$ and 1 and the same normalised weights.

Keep the evidence roles separate

Plan or input · Procedure · Permitted conclusion

Plan or input: $q=[1]$, keys 0 and $\ln(3)$

Procedure: Scaled scores 0, $\ln(3)$

Permitted conclusion: Softmax weights .25, .75

Plan or input: Values 2 and 6

Procedure: Weighted sum $.25*2 + .75*6$

Permitted conclusion: Output 5

Plan or input: Mask excludes B

Procedure: Only A permitted

Permitted conclusion: Weight 1 and output 2

Scores, normalised weights and value combinations are separate stages.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected learner request



AMBIGUOUS
Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING SOURCE

Evidence unavailable



ADVERSARIAL

Rules or trust challenged

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Set the draw to 0.97, keeping both distributions.

Prediction: Both temperatures select B.

Reason: 0.97 exceeds both A boundaries, 0.8 and approximately 0.9412.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

DECODING

method for choosing output
tokens

2

VARIABILITY

how much repeated outputs differ

3

TEMPERATURE INTUITION

setting that flattens or sharpens choices

4

SEEDED TEST

controlled way to compare runs

Word · My own example · Not this / boundary

Word: decoding

My own example: Selecting output tokens from a model's scores using a specified procedure.

Not this / boundary: _____

Word: variability

My own example: Differences observed across repeated runs, samples or conditions.

Not this / boundary: _____

Word: temperature intuition

My own example: A rough understanding of how temperature rescales token scores: higher values commonly spread choices more, without guaranteeing creativity or truth.

Not this / boundary: _____

Word: seeded test

My own example: A test containing a deliberately planted, known condition or error to check whether it is handled.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Sampling

Meaning: Selecting an alternative using a specified random procedure

Word: Greedy choice

Meaning: Selecting an alternative with largest next-token probability under a tie rule

Word: Temperature

Meaning: A positive parameter that changes the concentration of a distribution under a stated transformation

Word: Cumulative interval

Meaning: A range of draws assigned to an alternative using accumulated probability mass

Terms for A small attention calculation

Term · Meaning

Term: Scaled dot product

Meaning: A query-key dot product divided by the square root of key width

Term: Softmax

Meaning: Exponentiating scores and normalising their sum to obtain weights

Term: Score

Meaning: An unnormalised compatibility value before softmax

Term: Numerical stability

Meaning: An implementation choice that avoids avoidable arithmetic overflow or loss

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 4 • WORKED EVIDENCE

- 1 Low variability favors high-weight continuations repeatedly.
- 2 Higher variability gives lower-weight options more chances.
- 3 Diversity may help brainstorming but can increase off-task or unsupported detail.
- 4 Choose settings by task, then evaluate repeated runs—not one favorite output.

1. Low variability favors high-weight continuations repeatedly.
2. Higher variability gives lower-weight options more chances.
3. Diversity may help brainstorming but can increase off-task or unsupported detail.
4. Choose settings by task, then evaluate repeated runs—not one favorite output.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

The lower positive temperature concentrates this table on its largest entry. For $T=2$, use square roots before normalising; this flattens this unequal distribution. These changes leave the order of positive probabilities unchanged, so greedy still selects red. They change the sampling distribution, not the factual evidence behind a generated claim.

Our formula excludes $T=0$; dividing by zero is undefined. Some systems offer a zero-temperature setting as a separate greedy policy, but that is not a value to substitute into this formula. We also avoid claiming that any one setting removes all nondeterminism in an unspecified system.

Apply the connected idea: A small attention calculation

For $q=[1,1]$, $kA=[1,0]$ and $kB=[0,1]$, both dot products are 1. Divide by $\sqrt{2}$; equal scores give equal weights $1/2$ regardless of their common value. With values $[2,0]$ and $[0,4]$, the output is $[1,2]$. Do not omit the scale merely because it cancels in this tied example; unequal scores generally change concentration when scaled.

Apply masks before normalising over permitted scores. In the width-one example, excluding B leaves weight 1 for A and output 2. A key with the wrong width violates the dot-product contract. Values must share a width with one another, but their width need not equal the query/key width. An all-excluded mask is handled explicitly instead of taking a maximum of an empty list.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Sampling and Variability

A fictional two-token distribution illustrates variability in generation.

The facts we will use

At $T=1$, $A=0.8$ and $B=0.2$. For positive T use $p^{(1/T)}$, then normalise.

Use token order A,B and cumulative intervals $[0, \text{boundary})$, $[\text{boundary}, 1)$. Draw $u=0.85$.

Temperature-scaled toy probabilities

	T=1	T=0.5
A	0.8	0.9411764705882353
B	0.2	0.058823529411764705

Read a cell using both its row and column labels.

Sampling and Variability

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Original distribution

A boundary 0.8; $u=0.85$ selects B.



2. $T=0.5$ weights

Square probabilities: 0.64 and 0.04; total 0.68.



3. Normalise

$A=16/17 \approx 0.9412$; $B=1/17 \approx 0.0588$.



4. Same draw

0.85 is below $16/17$, so $T=0.5$ selects A.

CHANGE ONE THING

Set the draw to 0.97, keeping both distributions.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Original distribution

What happens and why: A boundary 0.8; $u=0.85$ selects B.

Step: 2. $T=0.5$ weights

What happens and why: Square probabilities: 0.64 and 0.04; total 0.68.

Step: 3. Normalise

What happens and why: $A=16/17 \approx 0.9412$; $B=1/17 \approx 0.0588$.

Step: 4. Same draw

What happens and why: 0.85 is below 16/17, so $T=0.5$ selects A.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Lower positive temperature concentrates this unequal distribution on its larger entry. It changes sampling probabilities, not evidence for whether A or B is true. Using the same draw isolates how the distribution changes the decision. This formula is a toy probability-space illustration corresponding to a temperature-scaled model distribution.

What this example does not establish

Two tokens and one draw do not predict every output; temperature alone does not solve hallucinations.

Read the labelled picture: Does $T=0.5$ make B impossible, and does choosing A guarantee factual accuracy?

Change one thing: Set the draw to 0.97, keeping both distributions. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: Higher variability makes a model more knowledgeable. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

A fair temperature comparison fixes the prefix, model distribution, candidate order and draw policy. Record multiple outcomes and evaluate task criteria separately. If the aim is creativity in fiction, variation may be useful; if the aim is reporting a sensor state, every factual claim still needs evidence. Never call a probability-concentration change an accuracy improvement without an appropriate evaluation.

Exit: a team lowers temperature and claims hallucinations are impossible. State what was changed and what evidence that claim still lacks.

Apply the connected idea: A small attention calculation

Check query/key dimensions, scale, permitted positions, finite scores, stable normalisation, weight sum and value widths in that order. A zero output can result from values cancelling; it does not imply zero attention weights. Record enough intermediate values for someone else to reproduce the trace.

Exit: two equal scores yield weights $1/2, 1/2$ and values 4,-4 yield output 0. Explain why zero output does not mean neither position was used.

8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Supported draws 0.2,0.6,0.9 at $T=1$ select red, blue, green. At $T=0.5$, 0.2 and 0.6 fall below $18/23$ and select red; 0.9 lies between $18/23$ and $45/46$ and selects blue. Keep exact fractions for boundaries, because rounding can misclassify a draw near an endpoint.

Guided: trace those three draws under both distributions and explain why the result is a procedure trace, not a frequency estimate from a large experiment.

Apply the connected idea: A small attention calculation

Supported scores 1000 and $1000 + \ln(3)$ would overflow an ordinary direct exponential calculation in many floating-point implementations. Maximum subtraction gives $-\ln(3)$ and 0, producing weights $1/4$ and $3/4$. Their common shift does not change the relative weights. This is a stability mechanism, not evidence about source correctness.

Guided: calculate the shifted scores and weights, then the output for values 4 and 8.

Practise with less help: Compute the adjusted table

Use workbook W2 for the connected example “Sampling and temperature”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Sampling and temperature. Proposed explanation: “Sampling means equal chances”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: A small attention calculation

Use its named workbook W2. Identify the given inputs, try the middle steps with less help, then explain your trace. Check this claim before group discussion: Scores already are probabilities

Explain the picture

Does $T=0.5$ make B impossible, and does choosing A guarantee factual accuracy?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → RUN → VERIFY → EXPLAIN

Run each setting five times on one fixed prompt, score consistency, usefulness, and unsupported detail, then justify a task-specific choice.

Run each setting five times on one fixed prompt, score consistency, usefulness, and unsupported detail, then justify a task-specific choice.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY
1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



VERIFIED STUDY BUDDY

Use low variability for structured study answers and allow more diversity only for labeled brainstorming.

THIS WEEK'S CHECKPOINT Use low variability for structured study answers and allow more diversity only for labeled brainstorming.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

Evidence for your project

Keep Study Buddy generation settings fixed during prompt comparisons and check evidence regardless of variability.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT CREATOR PROOF

Compare low- and high-variability story continuations.

CHALLENGE Compare low- and high-variability story continuations.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 4 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

One prompt produces three different valid quiz questions.

The question

Does variability mean failure?

Worked explanation

No. Sampling can vary outputs. Score correctness, relevance and format over repeated trials; consistency differs from quality.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Variability and quality are separate

Evidence or stage · Value or result

Evidence or stage: Trial A

Value or result: Different valid quiz question

Evidence or stage: Trial B

Value or result: Another valid quiz question

Evidence or stage: Trial C

Value or result: Check correctness, relevance and format

One prompt produces three different valid quiz questions.

Does variability mean failure?

No. Sampling can vary outputs. Score correctness, relevance and format over repeated trials; consistency differs from quality.

14. Try a new case independently

Independent case: Sample a new table independently

Probabilities $A=0.5, B=0.3, C=0.2$ in that order. Compute $T=0.5$ probabilities and cumulative boundaries. Trace $u=0.5$ and $u=0.9$ under $T=1$ and $T=0.5$. Explain the limit of the comparison.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 4 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Prompt A is run three times and returns three different stories. Does this alone prove the prompt changed? Describe a fair comparison of two prompts.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Use low variability for structured study answers and allow more diversity only for labeled brainstorming.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Prompt A gives three identical wrong answers. Prompt B gives two correct answers and one wrong answer. Compare consistency and correctness without declaring either generally superior.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional two-token distribution illustrates variability in generation.

By the end, I can · How I will show it

By the end, I can: Sample from a distribution using explicit cumulative intervals

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Calculate temperature-adjusted probabilities for a toy example

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Explain why random variation and factual reliability need separate evaluation

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Sample from a distribution using explicit cumulative intervals

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.